

Um Estudo Comparativo entre as Linguagens de Programação Java e Python



► História do Python - legibilidade, simplicidade e versatilidade



Surgimento

Python surgiu no ano de 1989, pelo programador e matemático Guido van Rossum



Propósito

Python foi pensando com três grandes propósitos legibilidade, simplicidade e versatilidade. Esses propósitos são os responsáveis por sua popularidade



Bibliotecas e Frameworks

Oferecem uma vasta gama de funcionalidades e soluções prontas para diversas aplicações em um código Python



Aplicações

As aplicações que utilizam Python e suas bibliotecas e framework são aplicações de ciência de dados, desenvolvimento web e IAs

► História do Java – “write once, run anywhere”



Surgimento

Java surgiu no ano de 1995, pelo cientista da computação James Gosling e sua equipe na empresa Sun Microsystems



Propósito

Java foi pensado para permitir a comunicação com qualquer aparelho eletrônico. Isso só é permitido graças a Java Virtual Machine (JVM)



Java Virtual Machine

A Java Virtual Machine, permite a portabilidade entre sistemas computacionais ao realizar o gerenciamento automático de memória e segurança



Aplicações

As aplicações que utilizam Java atualmente são sistemas de empresas, jogos e desenvolvimento web



► Bibliotecas Python

► Introdução das Linguagens

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► **Lógica e Sintaxe Java**

► Lógica e Sintaxe Java

- A principal característica dessa linguagem é o paradigma de **orientação a objetos**, portanto, todo código executado deve estar no escopo do método de um objeto instanciado, ou de uma classe quando esse for estático
- Java é uma tecnologia fortemente tipada, ou seja, a conversão de dados devem ser complexas e não flexíveis. É muito utilizado o casting explícito e implícito para objetos.

```
1 public class HelloWorld {  
2     public static void main(String[] args) {  
3         System.out.println("Hello, World!");  
4     }  
5 }
```

Categoria	Tipo	Tamanho
Inteiro	byte	8 bits
Inteiro	short	16 bits
Inteiro	int	32 bits
Inteiro	long	64 bits
Ponto Flutuante	float	32 bits
Ponto Flutuante	double	64 bits
Caracter	char	16 bits
Lógico	boolean	true / false

► Lógica e Sintaxe Java : outras regras

- Para criação de classes, deve seguir a estrutura :
tipo_de_acesso tipo_de_classe class nome_da_classe e
abertura de chaves.
- São usados construtores para instanciar objetos
- Metodos get e setters para o acesso de variáveis
- Letras maiúsculas diferem de letras minúsculas
- Todo programa deve haver uma classe main, com um método main estático, em que esse será chamado assim que o programa for executado
- Possui controles de fluxo(if, else, if else) e laços de repetição (for, while...)
- Conta com vetores e listas dinâmicas



► **Lógica e Sintaxe Python**



► Implementação Python

► Principais Implementações

01

Aplicações Web

02

Ciência de dados

03

Machine Learning

04

Manipulação de Imagens

► Aplicações Web



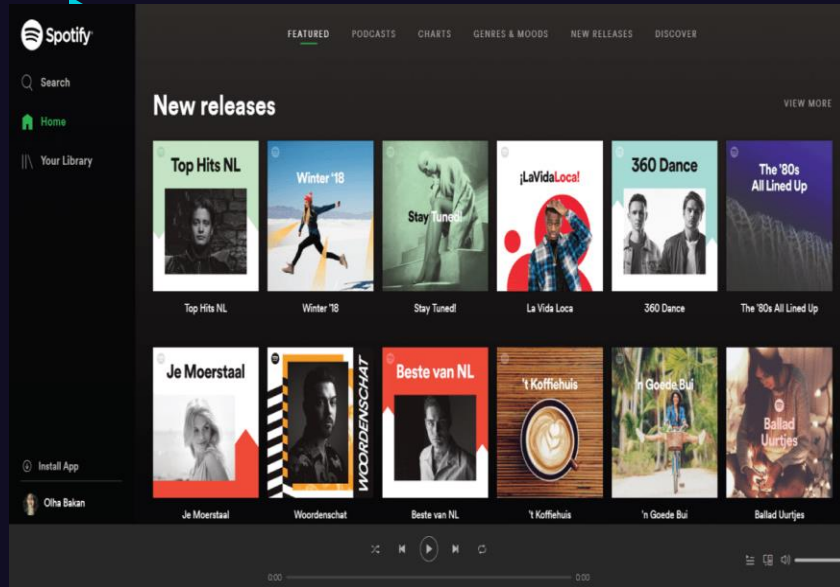
Aplicações Web

Python pode ser usado para desenvolvimento web com ajuda de alguns de seus frameworks, sendo os principais : Django e Flask. Aplicações como Youtube e Spotify usam dessa linguagem para gerar boa responsividade e experiência ao cliente



Ciência de dados

Na área de ciência de dados, essa linguagem é frequentemente usada, principalmente para manipular tabelas e grandes massas de dados, geralmente a partir do Excel. As bibliotecas mais populares utilizadas para essa área, são: Pandas (leitura de tabelas), Numpy (geração de gráficos) e Matplotlib (plotagem 2d)



Aplicação web com Django

```
1 golf_dict = {"ceu":["nuvem","nuvem","sol","chuva","sol","sol"],
2              "temperatura":[83,94,70,68,85,80],
3              "umidade":[86,65,90,75,None,90],
4              "vento":[None,True,False,False,False,True],
5              "joga":["sim","sim","sim","sim","nao","nao"]}
6 df = pd.DataFrame(golf_dict)
7 df
```

	ceu	temperatura	umidade	vento	joga
0	nuvem	83	86.0	None	sim
1	nuvem	94	65.0	True	sim
2	sol	70	90.0	False	sim
3	chuva	68	75.0	False	sim
4	sol	85	NaN	False	nao
5	sol	80	90.0	True	nao

Aplicação usando Pandas

► Principais Implementações

Machine Learning

Essa subárea da inteligência artificial consiste na ideia de fazer com que o código "aprenda" a lidar com o problema a partir de reconhecimento de padrões, treinados por ele. É um método extremamente atual e presente na realidade, assim, o Python consegue entregar boa performance e flexibilidade para execução. Principais bibliotecas usadas : TensorFlow, Keras e Sklearn (criação do algoritmo) e Numpy (gerar gráficos de acurácia)

Manipulação de imagens

Python é extremamente útil para trabalhar com imagens, desde a manipulação de seus pixels, até a realização de convoluções, ressaltando bordas, fundos ou texturas. O uso de ferramentas para essa manipulação são as próprias bibliotecas da linguagem, como : TensorFlow e OpenCv (processamento de imagens)

```
import pandas as pd
arquivo = pd.read_csv('C:/DT/wine_dataset.csv')
arquivo.head()

arquivo['style'] = arquivo['style'].replace('red', 0)
arquivo['style'] = arquivo['style'].replace('white', 1)

y = arquivo['style']
x = arquivo.drop('style', axis = 1)

from sklearn.model_selection import train_test_split
x_treino, x_teste, y_treino, y_teste = train_test_split(x, y, test_size = 0.3)

from sklearn.ensemble import ExtraTreesClassifier
modelo = ExtraTreesClassifier()
modelo.fit(x_treino, y_treino)

resultado = modelo.score(x_teste, y_teste)
print("Acurácia:", resultado)
```

Machine learning usando SkLearn e Pandas



Processamento de imagens e convolução

▶ WHAT IS A PITCH DECK?

A pitch deck is a short presentation that entrepreneurs use to describe their business and present it to potential investors. It usually consists of about 10 slides with:

- An overview of the company's mission and the product or service offered
- Market opportunity
- Team, financials, etc.

The goal is to garner enough interest from investors so that they are motivated to invest in the business



► MAIN IDEAS OF A PITCH DECK



PRESENTATION

A great presentation for a pitch deck should be concise, clear and easy to understand. Visual elements such as charts, graphs, images, and videos can help to make the presentation engaging



INVESTORS

When it comes to picking investors for a pitch deck, it's important to research potential investors and understand their preferences, goals and interests



VALUE

The value of a product is determined by how much customers are willing to pay for it. It's a combination of factors, including the quality of the product, its features, and how well it meets customer needs

► SIX RECOMMENDATIONS



KEEP IT SIMPLE

Structure your pitch deck clearly and make sure all the important points are easy to understand



USE VISUALS

Incorporate visuals or graphics to illustrate your message and keep the audience engaged



KEEP IT SHORT

A good pitch deck should be concise and clear; avoid trying to cram in too much information



LET'S MAKE IT

Include impactful stories, statistics, or facts that will help your audience remember your message afterwards



TEST & ITERATE

Practice presenting your deck beforehand with friends or colleagues for feedback, then use this input to refine it further



MAIN POINTS

A pitch deck is an overview of your business; focus on key points that get the most important ideas across



333,000

Users bought our product



9h 55m 23s

Estimated delivery time per unit



386,000 km

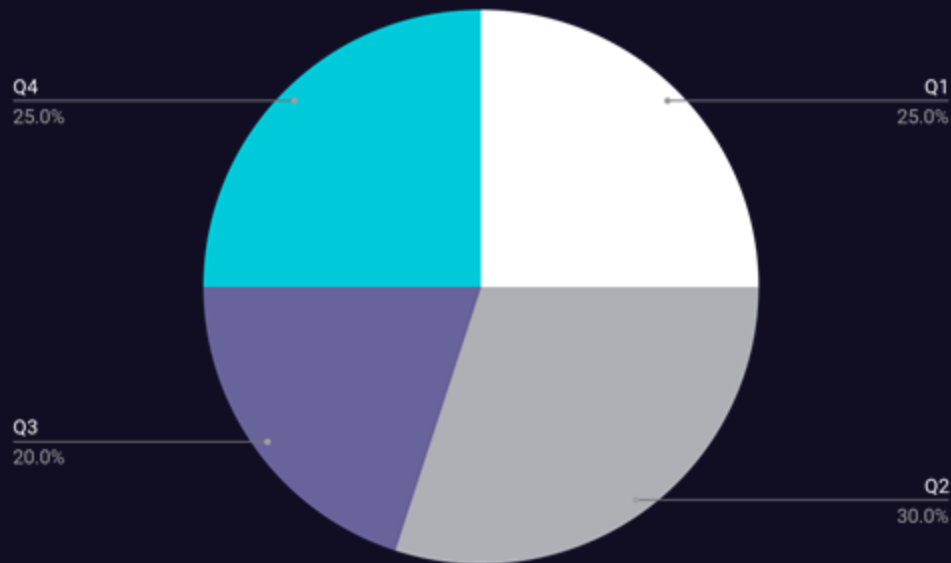
Avg. distance travelled by logistics team



► **4,498,300,000**

Number of users analyzed in our market research

► REVENUE BY QUARTER



Q1

January - March



Q2

April - June



Q3

July - September



Q4

October - December

Follow the link in the graph to modify its data and then paste the new one here. [For more info, click here](#)

► COMPETITION COMPARISON

	FEATURES	VALUE	PRICING	TRIAL	COMPETENCE	SHARE
Company A	Fuel economy	Special offers	\$23,000	No	Low	8%
Company B	Design	Reliability	\$27,000	No	Low	5%
Company C	Performance	Repairs	\$30,000	Yes	High	20%
Company D	Safety features	Marketing	\$24,000	No	High	22%
Company E	Technology	Customers	\$32,000	Yes	Low	10%
Company F	Comfort	Best prices	\$15,000	Yes	High	30%
Company G	Customization	Brand name	\$45,000	No	Low	5%

► PRODUCT INFOGRAPHIC

VISUALS

Showcase the design of your product



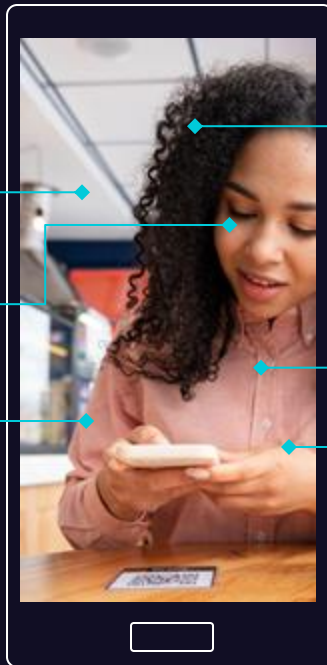
PRICE

Share the cost and price with investors



AVAILABILITY

When will the product be ready?



FEATURES

What makes your product unique?



USERS

Speak about the target audience



UPDATES

Do you plan on updating it?

► TIMELINE OF YOUR PRESENTATION



Provide a brief overview of the pitch deck's content



Engage the audience with a compelling introduction



Identify the customer's pain points and challenges



Describe how your product or service can solve the problem



Highlight the unique value proposition and benefits of your specific solution



Analyze the target market size, growth potential, and competition briefly

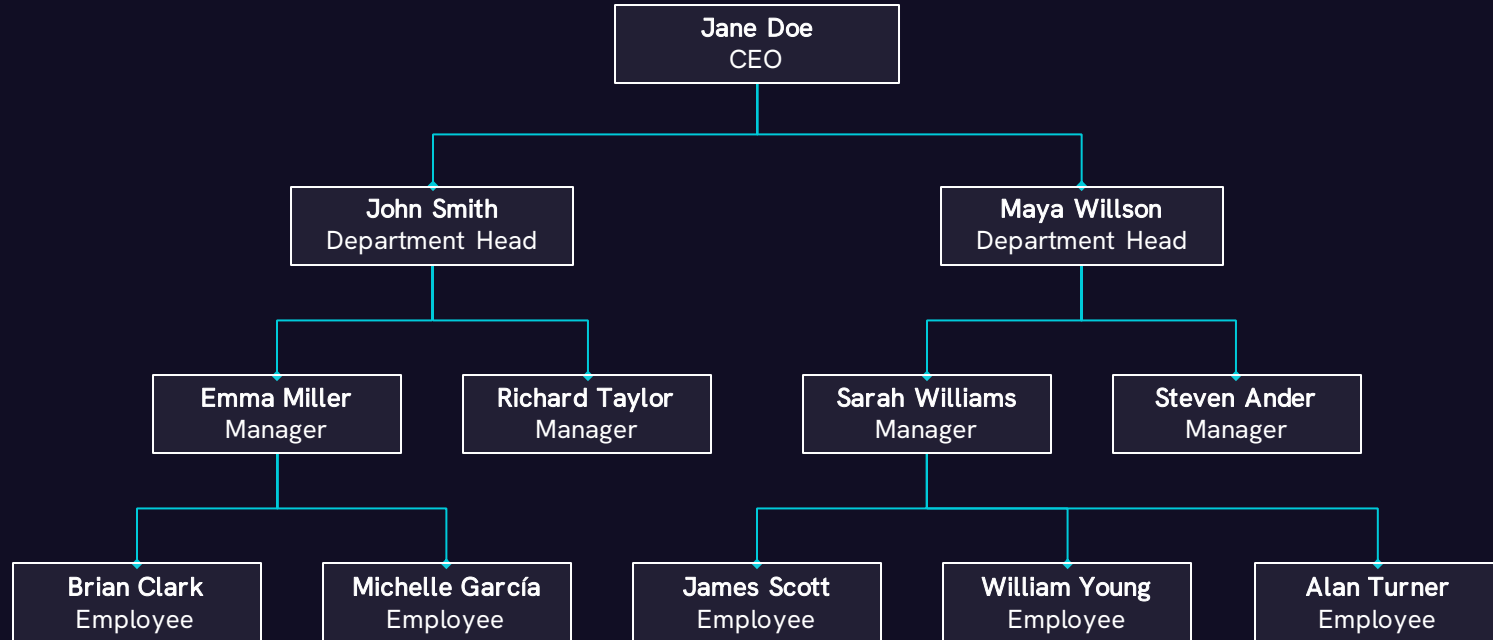


Explain your financial needs and briefly outline your funding allocation

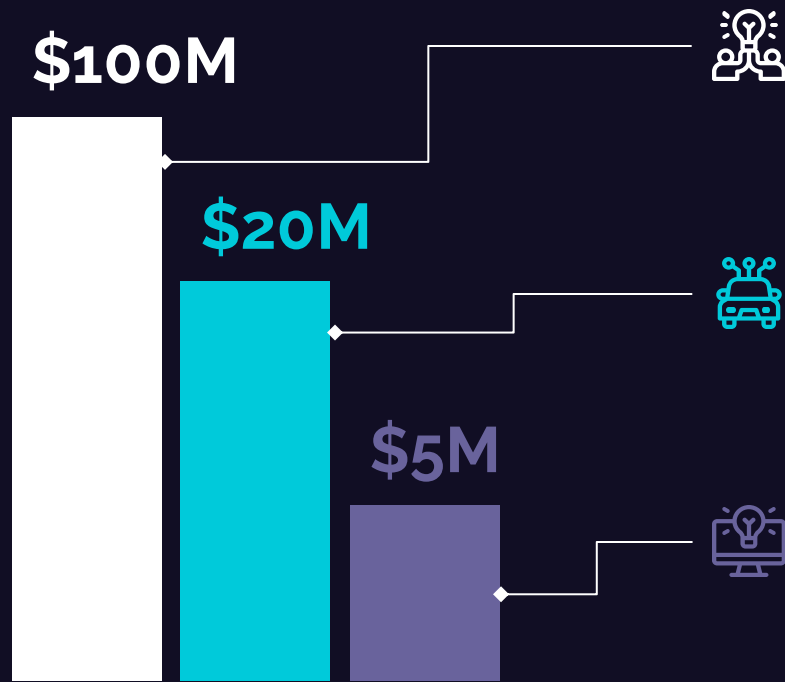


End with a clear and concise call to action

► ORGANIZATIONAL CHART



▶ MARKET SIZE OVERVIEW



OUTER CIRCLE

Include the total size of the market, which represents the entire potential customer base for the product or service



MIDDLE CIRCLE

Identify the target market for the product or service, which may be a subset of the total market. This could be based on factors such as demographics, geography, or specific needs



INNER CIRCLE

Indicate the current market size, which represents the portion of the target market that the company has successfully captured



► KPI DASHBOARD

PRODUCT	UNITS	REVENUE	RETURNS
Item 1	500	2,000,000	40
Item 2	1,000	50,750	10
Item 3	250	1,500,000	300
Item 4	500	2,000,000	40
Item 5	1,000	50,750	10
Item 6	250	1,500,000	300
Item 7	500	2,000,000	40
Item 8	1,000	50,750	10

\$4,000,000

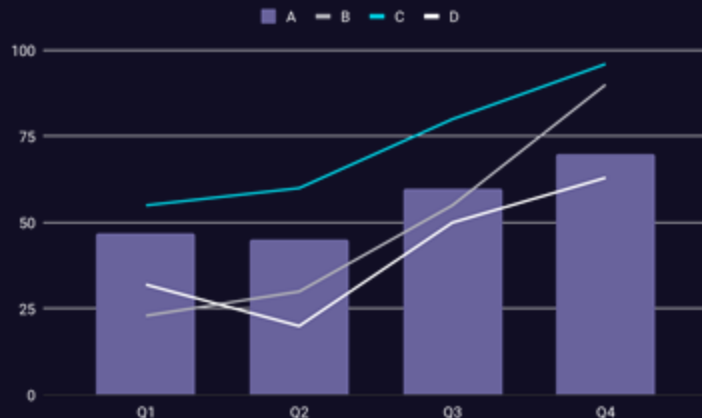
Revenue

\$100,000

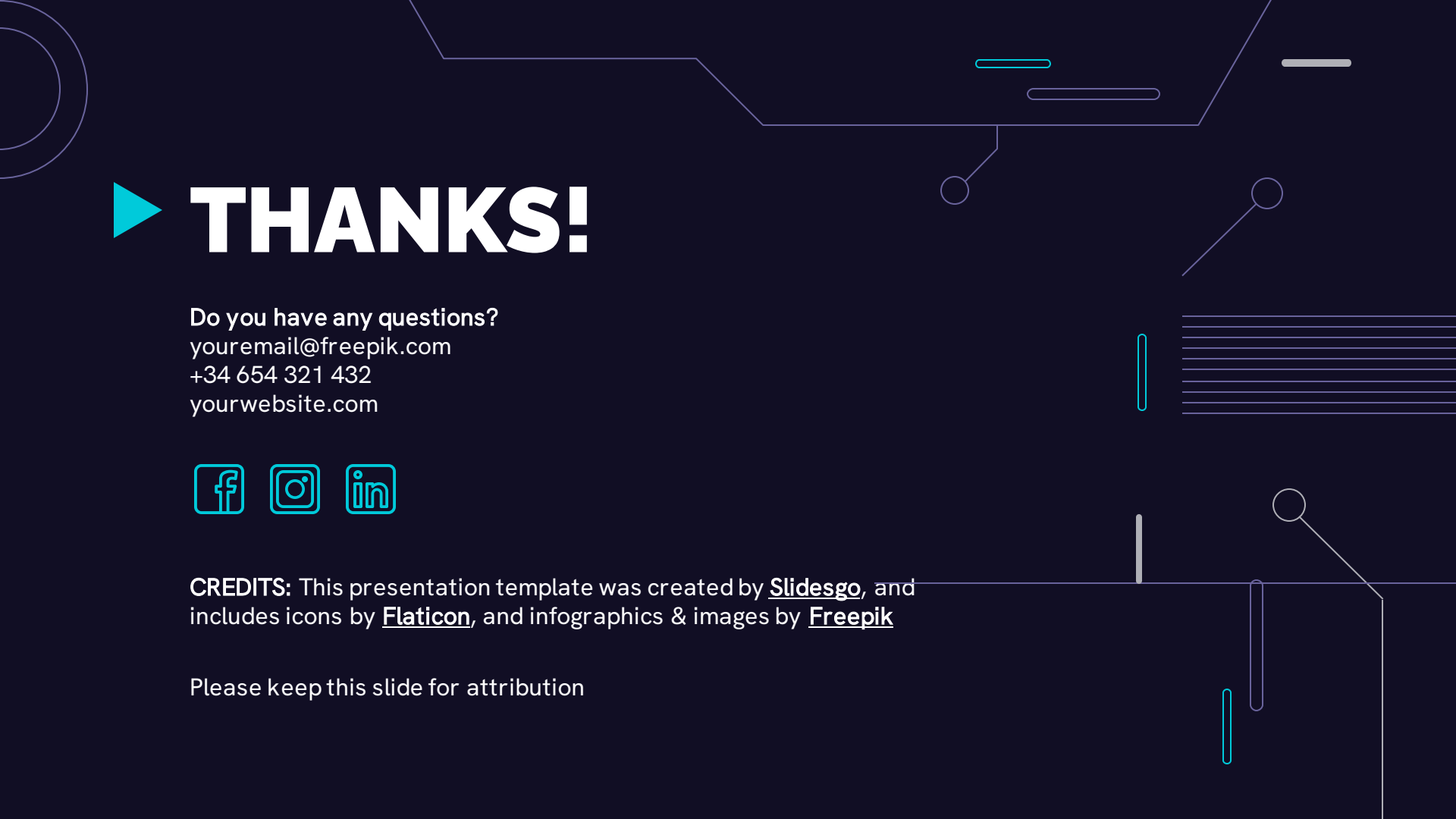
General costs

\$5,000

Benefit



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► THANKS!

Do you have any questions?

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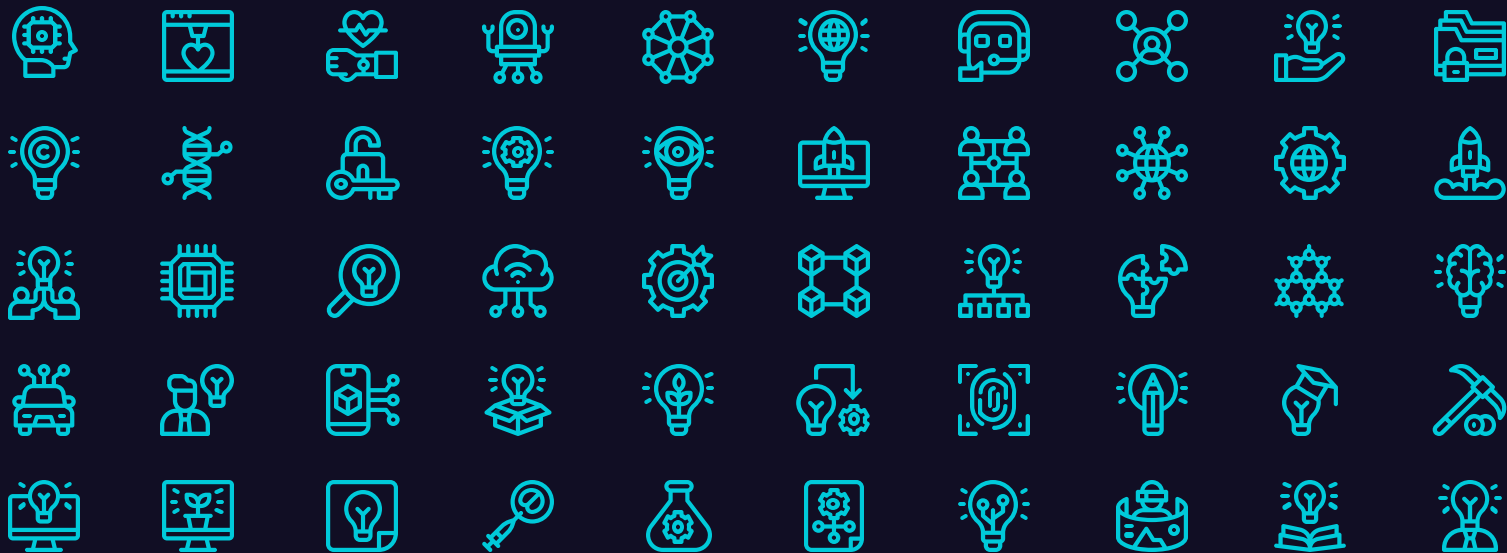
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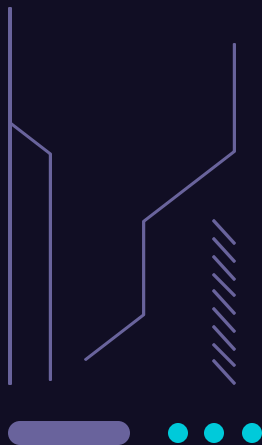
► ICON PACK



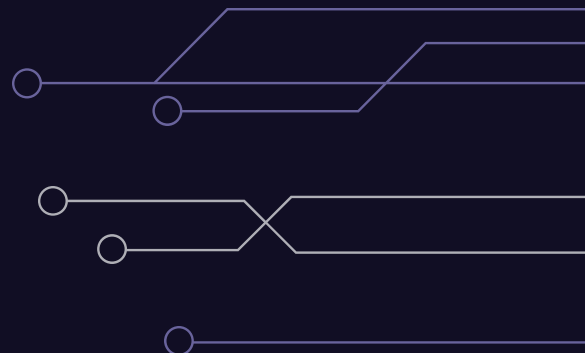
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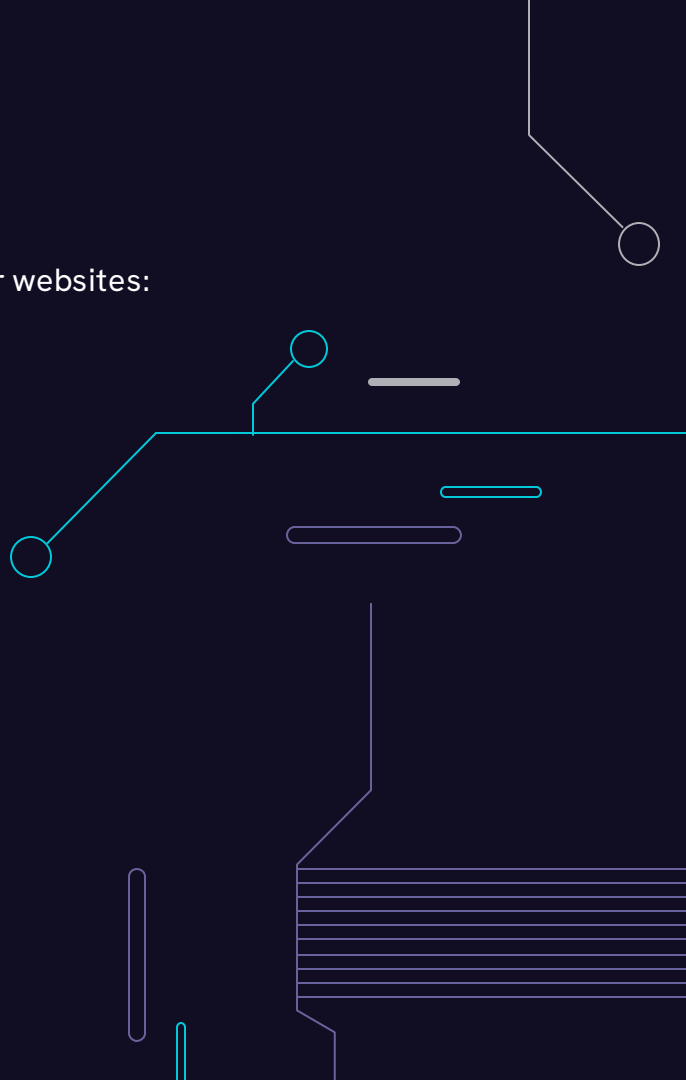
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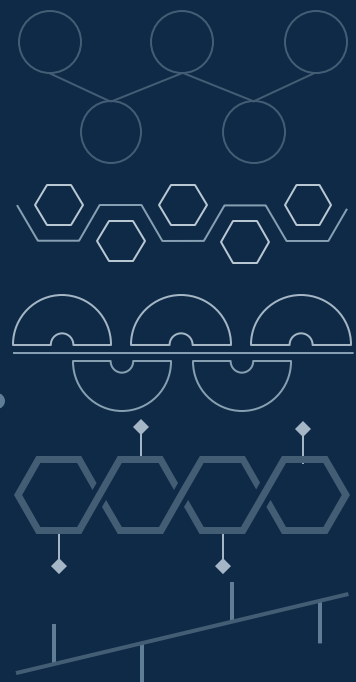
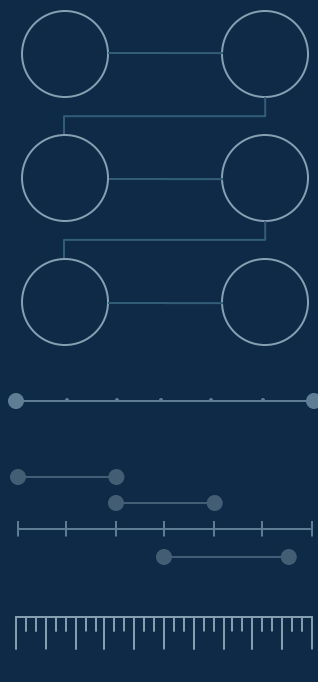
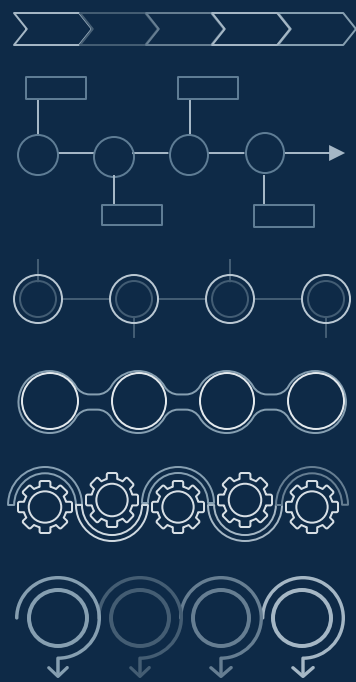
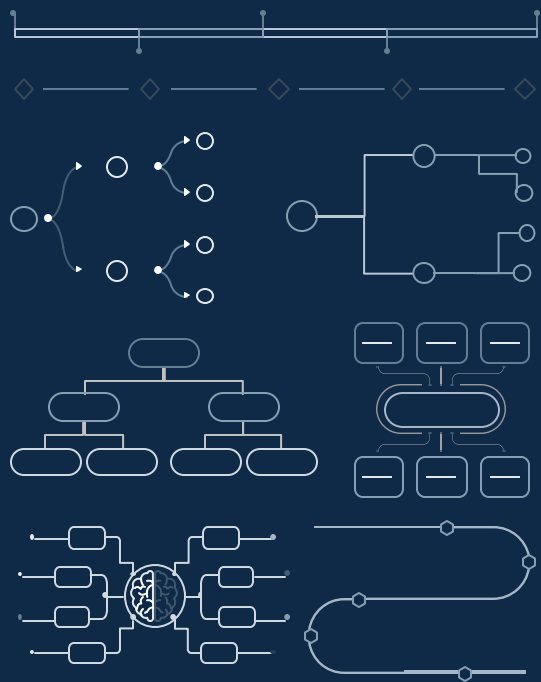
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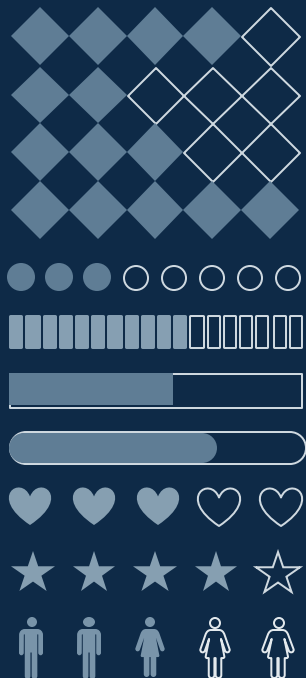
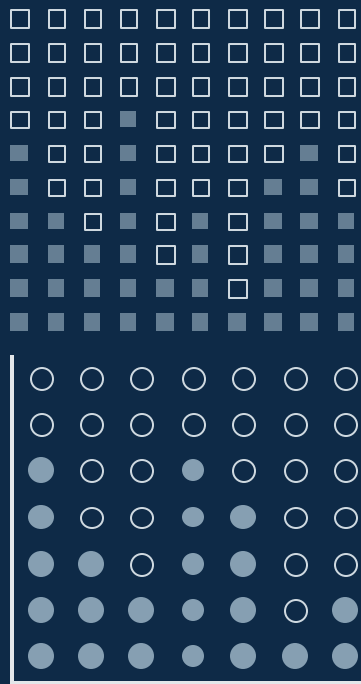




	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE
PHASE 1						
Task 1						
Task 2						
PHASE 2						
Task 1						
Task 2						

	JANUARY	FEBRUARY	MARCH	APRIL
PHASE 1				
Task 1				
Task 2				





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